

NARRATIVE REASONING

Evan Piermont
Royal Holloway, University of London
Department of Economics

Peio Zuazo-Garin
NYU Shanghai
Department of Economics

awAARe Conference
The Ninth Day of June in the Year of Our Lord Twenty Twenty Six



A **narrative** is a rhetorical device relating between distinct events:

- ◆ allows reasoners to make inferences

A **narrative** is a rhetorical device relating between distinct events:

- ◆ allows reasoners to make inferences
- ◆ For example, the narrative

“employers cannot afford to pay low-skilled labor more than they already do”

A **narrative** is a rhetorical device relating between distinct events:

- ◆ allows reasoners to make inferences
- ◆ For example, the narrative

“employers cannot afford to pay low-skilled labor more than they already do”

might allow a reasoner to infer that

“increasing the minimum wage” would “hurt low-skilled workers”

For a narrative to be **plausible** it must cohere with the reasoner's understanding: A reasoner

- ✦ who steadfastly believes that

“businesses drive down wages through monopsony power”

- ✦ will reject the narrative above, and as such,
- ✦ will draw different implications from a minimum wage increase.

The reasoner's understanding is based on limited / incomplete evidence

- ▶ some events / statements can be neither verified nor falsified
- ▶ his understanding is commensurately incomplete

This paper we

- (1) develop a theory of implication under limited evidence
- (2) define narrative plausibility
- (3) characterize the limits of narrative persuasion

Interestingly, the set of inferences a reasoner could be persuaded of is not necessarily transitively closed

Interestingly, the set of inferences a reasoner could be persuaded of is not necessarily transitively closed

In March 2026, the US announced it was lifting sanctions on Iranian oil.

The New York Times

War in the Middle East U.S.-Iran Talks Cease-Fire in Lebanon Strait of Hormuz Timeline of War

U.S. Lifts Sanctions on Some Iranian Oil to Get More to Market

Treasury Secretary Scott Bessent estimated that the move would add about 140 million barrels of crude to the oil market.

Listen · 2:52 min Share full article

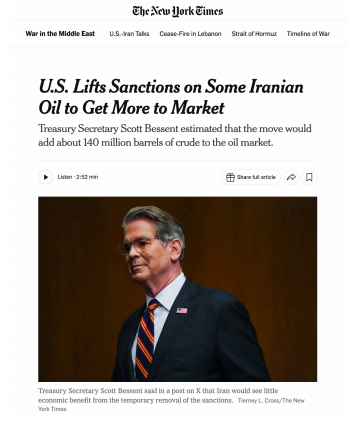


Treasury Secretary Scott Bessent said in a post on X that Iran would see little economic benefit from the temporary removal of the sanctions. Tierney L. Cross/The New York Times

Interestingly, the set of inferences a reasoner could be persuaded of is not necessarily transitively closed

In March 2026, the US announced it was lifting sanctions on Iranian oil.

This was justified via two narrative arguments.

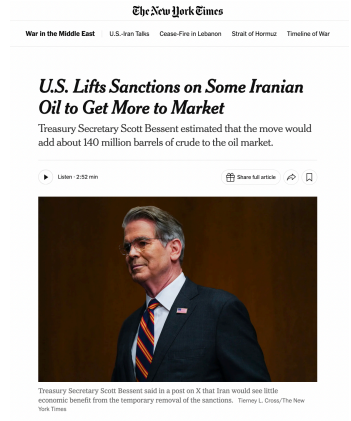


Interestingly, the set of inferences a reasoner could be persuaded of is not necessarily transitively closed

In March 2026, the US announced it was lifting sanctions on Iranian oil.

This was justified via two narrative arguments.

- (1) Lifting sanctions increases supply, decreasing the price of oil.

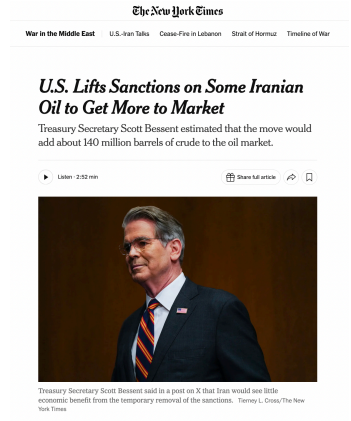


Interestingly, the set of inferences a reasoner could be persuaded of is not necessarily transitively closed

In March 2026, the US announced it was lifting sanctions on Iranian oil.

This was justified via two narrative arguments.

- (1) Lifting sanctions increases supply, decreasing the price of oil.
- (2) High oil prices help Iran, because it is a net exporter.



The causes and consequences of these two arguments can be arranged as:

a = "The US lift sanctions on Iranian Oil"

b = "The price of oil decreases"

c = "Iran is hurt"

➤ the first narrative attests the implication 'a implies b'

The causes and consequences of these two arguments can be arranged as:

a = "The US lift sanctions on Iranian Oil"

b = "The price of oil decreases"

c = "Iran is hurt"

- the first narrative attests the implication '**a** implies **b**'
- the second narrative attests the implication '**b** implies **c**.'

The causes and consequences of these two arguments can be arranged as:

a = “The US lift sanctions on Iranian Oil”

b = “The price of oil decreases”

c = “Iran is hurt”

- the first narrative attests the implication ‘**a** implies **b**’
- the second narrative attests the implication ‘**b** implies **c**.’
- One may be tempted then to conclude—as perhaps the US government inclined—that ‘**a** implies **c**.’

The causes and consequences of these two arguments can be arranged as:

a = “The US lift sanctions on Iranian Oil”

b = “The price of oil decreases”

c = “Iran is hurt”

- the first narrative attests the implication ‘**a** implies **b**’
 - ◆ predicated on the assumption that Iranian oil exports increase
- the second narrative attests the implication ‘**b** implies **c**.’
 - ◆ predicated on the assumption that oil exports do not increase
- One may be tempted then to conclude—as perhaps the US government inclined—that ‘**a** implies **c**.’

⇒ **STRINGENT**
IMPLICATION ⇐



➤ Traditional implication defines “a implies b” if:

whenever a is true it follows that b is true

- ▶ Traditional implication defines “a implies b” if:

whenever a is true it follows that b is true

- ▶ Regarding evidence:

- Traditional implication defines “a implies b” if:

whenever a is true it follows that b is true

- Regarding evidence:

(1) ‘*truth*’ becomes ‘*verification*’

- Traditional implication defines “a implies b” if:

whenever a is true it follows that b is true

- Regarding evidence:

(1) ‘*truth*’ becomes ‘*verification*’

(2) ‘*falsity*’ becomes ‘*falsification*’

- Traditional implication defines “a implies b” if:

whenever a is true it follows that b is true

- Regarding evidence:

- (1) ‘*truth*’ becomes ‘*verification*’
- (2) ‘*falsity*’ becomes ‘*falsification*’
- (3) new third option: neither

- Traditional implication defines “a implies b” if:

whenever a is true it follows that b is true

- Regarding evidence:

(1) ‘truth’ becomes ‘verification’

(2) ‘falsity’ becomes ‘falsification’

(3) new third option: neither

- To account for (3), we strengthen our notion of implication:

Stringent Implication

A statement a stringently implies b (written, $a \Rightarrow b$) if

whenever a cannot be falsified it follows that b can be verified

Consider the following events:

a = “The restaurant passed its health inspection”

b = “I am willing to eat at the restaurant”

- ❖ The restaurant can:
 - ❖ put a passing grade in the window (verifying **a**),
 - ❖ put a failing grade in the window (falsifying **a**), or,
 - ❖ have no sign at all.
- ❖ **a** does not stringently imply **b**, $a \not\Rightarrow b$
- ❖ A *lack of evidence* of failure is not sufficient to engender patronage.

Contrast this to the negated statements:

$\neg a$ = "The restaurant failed its health inspection"

$\neg b$ = "I am unwilling to eat at the restaurant"

- ◆ Here, $\neg a \Rightarrow \neg b$
- ◆ The lack of a passing grade is itself grounds for dismissal

To elicit stringent implication, $\Rightarrow$, ask the reasoner to choose between:

Option 1

Win \$10 if

- ▶ provide falsification of **a**, or,
- ▶ provide verification of **b**

Option 2

Win \$10 for sure

We then define **a** $\Rightarrow$ **b** via indifference between the options

Let A be a set of statements (descriptions of events):

Evidential Structure

An **evidential structure** is a triple (W, v, f) where:

- ❖ W is a set of states
- ❖ $v : A \rightarrow 2^W$ is a verification assignment, and,
- ❖ $f : A \rightarrow 2^W$ is a falsification assignment.

Let A be a set of statements (descriptions of events):

Evidential Structure

An **evidential structure** is a triple (W, v, f) where:

- ✦ W is a set of states
- ✦ $v : A \rightarrow 2^W$ is a verification assignment, and,
- ✦ $f : A \rightarrow 2^W$ is a falsification assignment.

Call an evidential structure

- ✦ **coherent** if $v(a) \cap f(a) = \emptyset$ for all a , and,
- ✦ **complete** if it is coherent and $v(a) \cup f(a) = W$ for all a .

For an evidential structure (W, v, f) , let $f^c : a \mapsto (W - f(a))$

Representation

An evidential structure (W, v, f) **represents** $\Rightarrow$ for all $a, b \in A$

$$a \Rightarrow b \quad \text{if and only if} \quad f^c(a) \subseteq v(b)$$

For an evidential structure (W, v, f) , let $f^c : a \mapsto (W - f(a))$

Representation

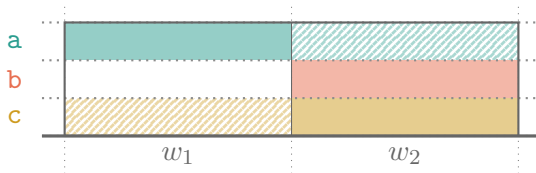
An evidential structure (W, v, f) **represents** $\Rightarrow$ for all $a, b \in A$

$$a \Rightarrow b \quad \text{if and only if} \quad f^c(a) \subseteq v(b)$$

whenever a cannot be falsified it follows that b can be verified

Let $A = \{a, b, c\}$ and take $\Rightarrow$ as:

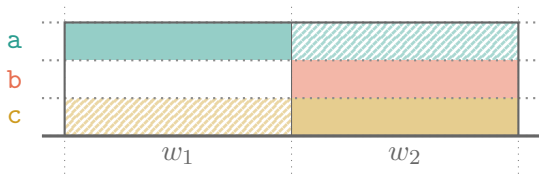
	a	b	c
a	$a \Rightarrow a$		
b			$c \Rightarrow b$
c			$c \Rightarrow c$



♦ Solid cells are verification and hashed cells are falsification

Let $A = \{a, b, c\}$ and take $\Rightarrow$ as:

	a	b	c
a	$a \Rightarrow a$		
b			$c \Rightarrow b$
c			$c \Rightarrow c$



- ◆ Solid cells are verification and hashed cells are falsification
- ◆ Notice, $b \not\Rightarrow b$, since $f^c(b) = \{w_1, w_2\} \not\subseteq \{w_2\} = v(b)$

What axioms on $\Rightarrow$ ensure that a representation exists?

What axioms on $\Rightarrow$ ensure that a representation exists?

Theorem

Every binary relation $\Rightarrow$ can be represented by an evidential structure.

Moreover,

- ▶ $\Rightarrow$ is transitive iff represented by a *coherent* evidential structure
- ▶ $\Rightarrow$ is a partial order iff represented by a *complete* evidential structure

PROOF. Let $W = A$ and

$$v : a \mapsto \{b \mid b \Rightarrow a\}$$

$$f : a \mapsto \{b \mid \exists c, a \Rightarrow c, b \not\Rightarrow c\}$$



This canonical representation satisfies a “tightness condition” (see paper)



NARRATIVES



Narratives

A **narrative** $\mathbf{N}$ is a reflexive and transitive relation on A

Narratives

A **narrative** $\mathbf{N}$ is a reflexive and transitive relation on A

By the prior theorem narratives can be represented by

- ♦ *complete* evidential structures (where $v(\mathbf{a}) = f^c(\mathbf{a})$)

Narratives

A **narrative** $\mathbf{N}$ is a reflexive and transitive relation on A

By the prior theorem narratives can be represented by

- ♦ *complete* evidential structures (where $v(\mathbf{a}) = f^c(\mathbf{a})$)
- ♦ Define $t(\mathbf{a}) := v(\mathbf{a}) = f^c(\mathbf{a})$; for all $\mathbf{a}, \mathbf{b} \in A$

$$\mathbf{a} \mathbf{N} \mathbf{b} \quad \text{if and only if} \quad t(\mathbf{a}) \subseteq t(\mathbf{b})$$

Plausibility

A narrative $\mathbf{N}$ is **plausible** for $\Rightarrow$, if there exists some (W, v, f, t) such that for all $a, b \in A$

Plausibility

A narrative $\mathbf{N}$ is **plausible** for $\Rightarrow$, if there exists some (W, v, f, t) such that for all $a, b \in A$

(P1) $a \Rightarrow b$ if and only if $f^c(a) \subseteq v(b)$

Plausibility

A narrative $\mathbf{N}$ is **plausible** for $\Rightarrow$, if there exists some (W, v, f, t) such that for all $a, b \in A$

(P1) $a \Rightarrow b$ if and only if $f^c(a) \subseteq v(b)$

(P2) $a \mathbf{N} b$ if and only if $t(a) \subseteq t(b)$, and,

Plausibility

A narrative $\mathbf{N}$ is **plausible** for $\Rightarrow$, if there exists some (W, v, f, t) such that for all $a, b \in A$

(P1) $a \Rightarrow b$ if and only if $f^c(a) \subseteq v(b)$

(P2) $a \mathbf{N} b$ if and only if $t(a) \subseteq t(b)$, and,

(P3) $v(a) \subseteq t(a) \subseteq f^c(a)$.

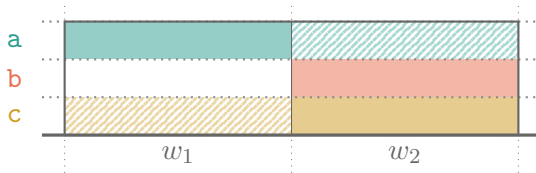
Recall $\Rightarrow$ defined over

a = "The US lift sanctions on Iranian Oil"

b = "The price of oil decreases"

c = "Iran is hurt"

	a	b	c
a	$a \Rightarrow a$		
b			$c \Rightarrow b$
c			$c \Rightarrow c$



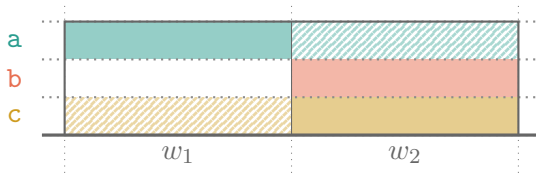
Recall $\Rightarrow$ defined over

a = "The US lift sanctions on Iranian Oil"

b = "The price of oil decreases"

c = "Iran is hurt"

	a	b	c
a	$a \Rightarrow a$		
b			$c \Rightarrow b$
c			$c \Rightarrow c$



There are two plausible narratives:

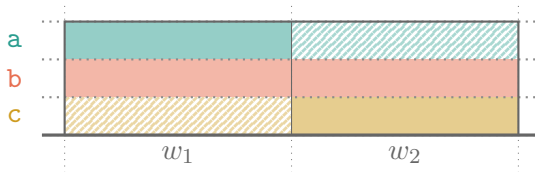
Recall $\Rightarrow$ defined over

a = "The US lift sanctions on Iranian Oil"

b = "The price of oil decreases"

c = "Iran is hurt"

	a	b	c
a	a N_1 a		
b	a N_1 b	b N_1 b	c N_1 b
c			c N_1 c



There are two plausible narratives:

(1) Iran increases exports, **a** N_1 **b**, but by revealed preference **b** N_1 **c**

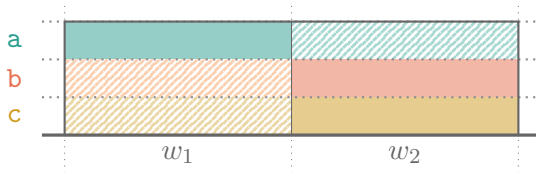
Recall $\Rightarrow$ defined over

a = "The US lift sanctions on Iranian Oil"

b = "The price of oil decreases"

c = "Iran is hurt"

	a	b	c
a	aN_2a		
b		bN_2b	cN_2b
c		bN_2c	cN_2c



There are two plausible narratives:

- (1) Iran increases exports, aN_1b , but by revealed preference bN_1c
- (2) Lower oil prices would hurt Iran, bN_2c , so does not increase exports aN_2b



→ **LENIENT**
IMPLICATION ←



Lenient Implication

Call $\rightarrow$ the **leniency** of $\Rightarrow$ if:

$a \rightarrow b$ if and only if $c \Rightarrow d$ whenever $c \Rightarrow a$ and $b \Rightarrow d$

Lenient Implication

Call $\rightarrow$ the **leniency** of $\Rightarrow$ if:

$a \rightarrow b$ if and only if $c \Rightarrow d$ whenever $c \Rightarrow a$ and $b \Rightarrow d$

► If $\Rightarrow$ is transitive, then $\rightarrow$ is reflexive.

Lenient Implication

Call $\rightarrow$ the **leniency** of $\Rightarrow$ if:

$a \rightarrow b$ if and only if $c \Rightarrow d$ whenever $c \Rightarrow a$ and $b \Rightarrow d$

- ▶ If $\Rightarrow$ is transitive, then $\rightarrow$ is reflexive.
- ▶ If $\Rightarrow$ is reflexive, then $\rightarrow$ is transitive.

Lenient Implication

Call $\rightarrow$ the **leniency** of $\Rightarrow$ if:

$a \rightarrow b$ if and only if $c \Rightarrow d$ whenever $c \Rightarrow a$ and $b \Rightarrow d$

- ▶ If $\Rightarrow$ is transitive, then $\rightarrow$ is reflexive.
- ▶ If $\Rightarrow$ is reflexive, then $\rightarrow$ is transitive.
- ▶ If $\Rightarrow$ is transitive, then $\Rightarrow \subseteq \rightarrow$.

Lenient Implication

Call $\rightarrow$ the **leniency** of $\Rightarrow$ if:

$a \rightarrow b$ if and only if $c \Rightarrow d$ whenever $c \Rightarrow a$ and $b \Rightarrow d$

- ▶ If $\Rightarrow$ is transitive, then $\rightarrow$ is reflexive.
- ▶ If $\Rightarrow$ is reflexive, then $\rightarrow$ is transitive.
- ▶ If $\Rightarrow$ is transitive, then $\Rightarrow \subseteq \rightarrow$.
- ▶ If $\Rightarrow$ is reflexive, then $\rightarrow \subseteq \Rightarrow$.

Let $A = \{a, b, c\}$ and take $\Rightarrow$ as:

	a	b	c
a	$a \Rightarrow a$		
b			$c \Rightarrow b$
c			$c \Rightarrow c$

✦ Transitive, not reflexive

Then the leniency of $\Rightarrow$ is:

	a	b	c
a	$a \rightarrow a$		
b	$a \rightarrow b$	$b \rightarrow b$	$c \rightarrow b$
c		$b \rightarrow c$	$c \rightarrow c$

✦ Reflexive, not transitive

Why is this called a *leniency*?

Why is this called a *leniency*?

Theorem

Fix $\Rightarrow$. The following are equivalent:

- (1) $\rightarrow$ is the leniency of $\Rightarrow$,
- (2) $a \rightarrow b$ iff for the canonical representation (W, v, f) , $v(a) \subseteq f^c(b)$
- (3) $a \rightarrow b$ iff for some representation $(\bar{W}, \bar{v}, \bar{f})$, $\bar{v}(a) \subseteq \bar{f}^c(b)$.

Why is this called a *leniency*?

Theorem

Fix $\Rightarrow$. The following are equivalent:

- (1) $\rightarrow$ is the leniency of $\Rightarrow$,
- (2) $a \rightarrow b$ iff for the canonical representation (W, v, f) , $v(a) \subseteq f^c(b)$
- (3) $a \rightarrow b$ iff for some representation $(\bar{W}, \bar{v}, \bar{f})$, $\bar{v}(a) \subseteq \bar{f}^c(b)$.

whenever a can be verified it follows that b cannot be falsified

Theorem

Fix $\Rightarrow$ and let $\rightarrow$ be its leniency. Then a narrative $\mathbf{N}$ is plausible if and only if

$$\Rightarrow \subseteq \mathbf{N} \subseteq \rightarrow$$

✦ That is,

$a \Rightarrow b$ implies $a\mathbf{N}b$ and $a\mathbf{N}b$ implies $a \rightarrow b$.

$\Rightarrow$	a	b	c
a	$a \Rightarrow a$		
b			$c \Rightarrow b$
c			$c \Rightarrow c$

N_1	a	b	c
a	aN_1a		
b	aN_1b	bN_1b	cN_1b
c			cN_1c

N_2	a	b	c
a	aN_2a		
b		bN_2b	cN_2b
c		bN_2c	cN_2c

$\rightarrow$	a	b	c
a	$a \rightarrow a$		
b	$a \rightarrow b$	$b \rightarrow b$	$c \rightarrow b$
c		$b \rightarrow c$	$c \rightarrow c$

$\Rightarrow$	a	b	c
a	$a \Rightarrow a$		
b			$c \Rightarrow b$
c			$c \Rightarrow c$

N_1	a	b	c
a	aN_1a		
b	aN_1b	bN_1b	cN_1b
c			cN_1c

N_2	a	b	c
a	aN_2a		
b		bN_2b	cN_2b
c		bN_2c	cN_2c

$\rightarrow$	a	b	c
a	$a \rightarrow a$		
b	$a \rightarrow b$	$b \rightarrow b$	$c \rightarrow b$
c		$b \rightarrow c$	$c \rightarrow c$

$\Rightarrow$	a	b	c
a	$a \Rightarrow a$		
b			$c \Rightarrow b$
c			$c \Rightarrow c$

N_1	a	b	c
a	aN_1a		
b	aN_1b	bN_1b	cN_1b
c			cN_1c

N_2	a	b	c
a	aN_2a		
b		bN_2b	cN_2b
c		bN_2c	cN_2c

$\rightarrow$	a	b	c
a	$a \rightarrow a$		
b	$a \rightarrow b$	$b \rightarrow b$	$c \rightarrow b$
c		$b \rightarrow c$	$c \rightarrow c$



THANK YOU

(all proof verified in lean)

